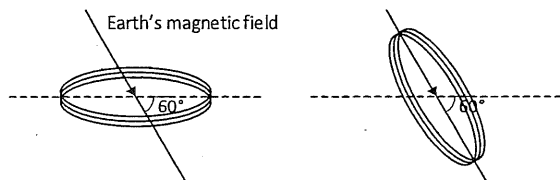


5. [HCI/H2/Prelims 2018/P1/23 Modified]

A coil of wire of 3 turns and cross sectional area 0.30 m^2 is placed on the ground. Earth's magnetic field of 0.045 T is directed at an angle of 60° below horizontal as shown.

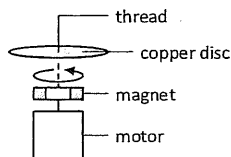


What is the magnitude of the change in magnetic flux linkage through the coil when the coil is rotated such that it is now parallel to the Earth's magnetic field?

- A 0.020 Wb B 0.035 Wb C 0.040 Wb D 0.070 Wb

6. [HCI/H2/Prelims 2018/P1/24 Modified]

A copper disc is suspended freely from the ceiling and is positioned directly above a rotating magnet.

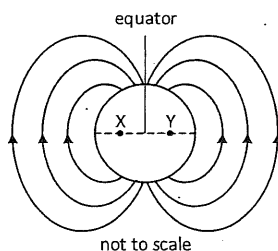


Which of the following statements is correct?

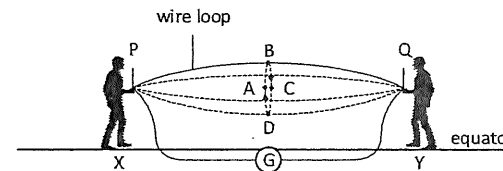
- A The disc remains stationary as copper is not magnetic.
 B The disc rotates in the same direction as the magnet as copper is magnetic.
 C The disc rotates in the same direction as the magnet as eddy currents are induced in the disc.
 D The disc rotates in the opposite direction as the magnet as eddy currents are induced in the disc.

7. [IJC/H2/Prelims 2018/P1/24 Modified]

The figure shows the magnetic field around the Earth.



Two students standing a few metres apart on the equator at points X and Y, where the Earth's magnetic field is parallel to the ground and acting into the page, hold a loop of copper wire between them. Part of the loop is rotated like a skipping rope as shown, while the other part remains motionless on the ground.

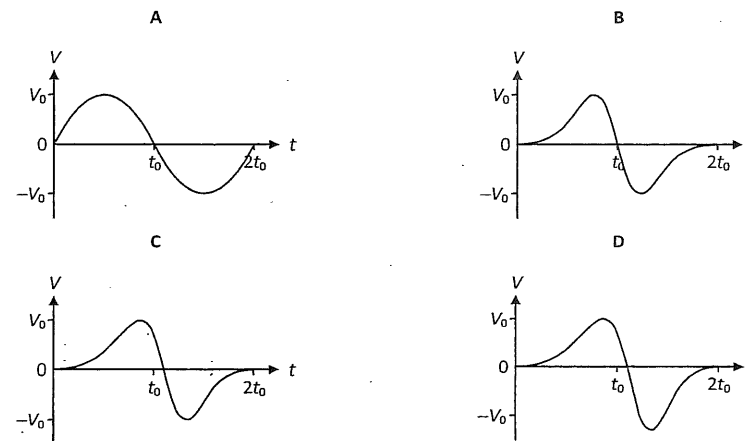


At which point A, B, C and D during the rotation of the wire does the maximum current flow in the direction from P to Q through the moving part of the wire?

8. [IJC/H2/Prelims 2018/P1/25 Modified]

A bar magnet is dropped from rest in a vertical solenoid that is connected to a sensitive voltage sensor. The magnet spent about duration of $2t_0$ inside the solenoid.

Which of the following graphs best represents the variation with time t of the voltage V recorded by the sensor?



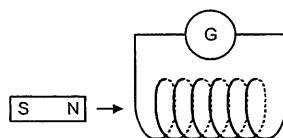
9. [MI/H2/Prelims 2018/P1/26 Modified]

An e.m.f. is induced in a coil of wire that is rotating about a vertical axis inside a uniform horizontal magnetic field. Which one of the following does not affect the magnitude of e.m.f. induced in the coil?

- A the angular velocity of the coil
 B the resistance of the coil
 C the number of turns of the coil
 D the magnetic flux density

10. [MJC/H2/Prelims 2018/P1/24]

The North pole of a bar magnet is pushed into the end of a coil of wire. The maximum movement of the galvanometer needle is 10 units to the left.

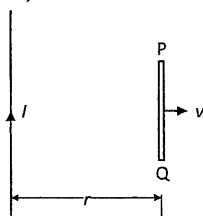


The South pole of the magnet is then pushed into the other end of the coil at half the speed. What is the maximum movement of the galvanometer needle?

- A less than 10 units to the left
- B less than 10 units to the right
- C more than 10 units to the left
- D more than 10 units to the right

11. [NYJC/H2/Prelims 2018/P1/22]

The diagram shows a wire carrying a current I and a straight conductor PQ is placed on the same vertical plane as the wire. PQ is moved at constant speed v away from the wire.

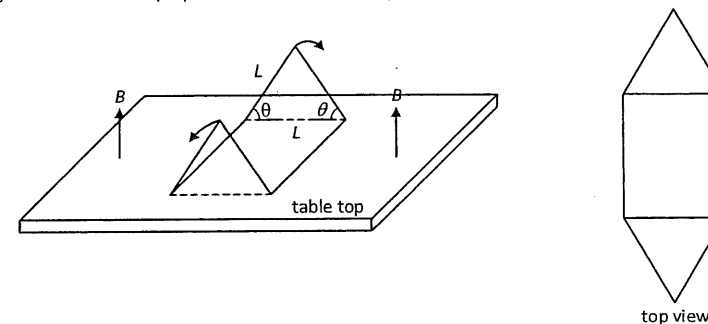


How will the magnitude of the induced e.m.f. in PQ vary and which end will be at a higher potential?

	magnitude of induced e.m.f.	end at higher potential
A	decrease	P
B	decrease	Q
C	increase	P
C	increase	Q

12. [NYJC/H2/Prelims 2018/P1/23]

The wire shown in the figure below is bent into the shape of a tent, with $\theta = 60^\circ$ and $L = 1.5$ m, and placed in a 0.30 T magnetic field directed perpendicular to the table-top.



If the tent is flattened on the table in 0.10 s as shown (top view), what is the average e.m.f. induced in the wire during this time?

- A 2.92 V
- B 5.85 V
- C 7.15 V
- D 9.21 V

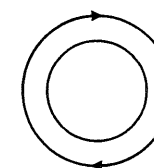
13. [NYJC/H2/Prelims 2018/P1/23]

A circular loop of wire of radius 0.20 m is placed in a uniform magnetic field of 1.2 T that is normal to the plane of the loop. The loop is rotated about an axis along a diameter at a rate of 50 revolutions per minute. What is the maximum e.m.f. induced in the coil?

- A 0.13 V
- B 0.15 V
- C 0.79 V
- D 47 V

14. [NYJC/H2/Prelims 2018/P1/24]

A small circular coil lies within a large circular coil as shown. The two coils are horizontal and concentric. The large coil carries a constant current in a clockwise direction.

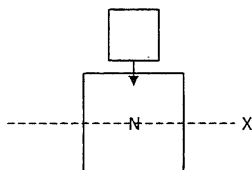


The smaller coil is pulled perpendicularly away from the plane of the paper. Which of the following describes the direction of the induced current in the small coil and the resultant force between the coils?

	induced current	force between coils
A	clockwise	attraction
B	clockwise	repulsion
C	anticlockwise	attraction
C	anticlockwise	repulsion

15. [PJC/H2/Prelims 2018/P1/22]

A square loop of copper wire is held above and near to a pole of a strong magnet as shown below.

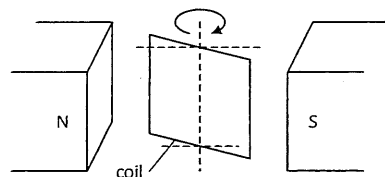


Which statement about the acceleration of the loop is correct after the loop is released?

- A The acceleration is greater than g just above X and less than g just below X
- B The acceleration is less than g just above X and greater than g just below X.
- C The acceleration is greater than g both just above and below X.
- D The acceleration is less than g both just above and below X.

16. [PJC/H2/Prelims 2018/P1/23]

A rectangular coil of wire is rotated at a constant angular speed in a magnetic field as shown below. An e.m.f. of peak value E_0 is generated. The coil is now rotated at twice the original frequency in a magnetic field whose strength is one-third of the original.

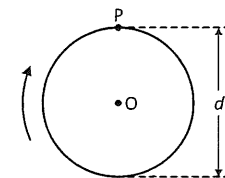


What is the peak e.m.f. generated now?

- A $6E_0$
- B $5E_0$
- C $\frac{3}{2}E_0$
- D $\frac{2}{3}E_0$

17. [RVHS/H2/Prelims 2018/P1/25]

An aluminum disc of diameter d rotates about its centre O. A uniform magnetic field B acts perpendicularly to the surface of the disc. A point P on the rim moves at a constant speed v . A steady electromotive force E is generated between O and P.



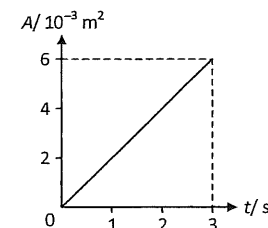
What is the expression for E ?

- A $\frac{1}{4}Bdv$
- B $\frac{1}{2}Bdv$
- C Bdv
- D Bv

18. [TPJC/H2/Prelims 2018/P1/27]

A single circular loop of wire moves in a uniform magnetic field of flux density 1.2 T.

The graph shows how the area A of the loop that is perpendicular to the magnetic field varies with time.



What is the e.m.f. induced in the loop?

- A $1.2 \times 10^{-3} \text{ V}$
- B $2.4 \times 10^{-3} \text{ V}$
- C $3.6 \times 10^{-3} \text{ V}$
- D $7.2 \times 10^{-3} \text{ V}$

19. [VJC/H2/Prelims 2018/P1/25]

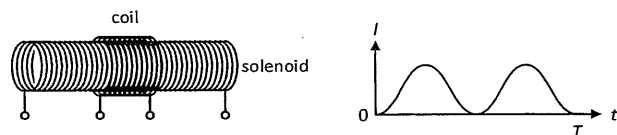
A copper disc of area A rotates at angular frequency ω at the centre of a long solenoid of length L and N turns. The solenoid carries a current I . The plane of the disc is normal to the magnetic flux density. The rate of rotation is adjusted so that the e.m.f. generated between the centre of the disc and its rim is 10% of the potential difference across the ends of the solenoid.

Which expression gives the potential difference across the ends of the solenoid?

- A $0.016\mu_0 NIA\omega/L$
- B $1.6\mu_0 NIA\omega/L$
- C $0.1\mu_0 NIA\omega$
- D $10\mu_0 NIA\omega$

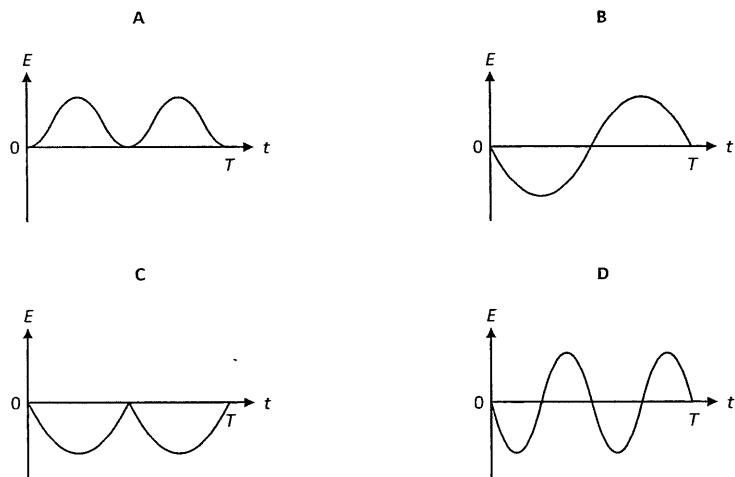
20. [VJC/H2/Prelims 2018/P1/26]

The figure below shows a short coil wound over the middle of a long solenoid.



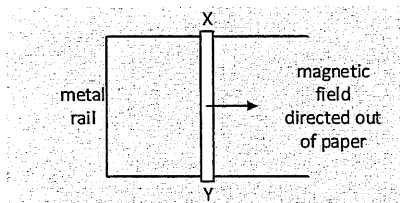
The solenoid current I is varied with time t as shown in the graph. As a consequence, the flux density of the magnetic field due to the solenoid varies with time.

Which graph shows the variation of induced e.m.f. E in the short coil with time t ?



21. [YJC/H2/Prelims 2018/P1/27]

The figure shows a metal rod XY moving to the right on a metal rail, perpendicular to a magnetic field.

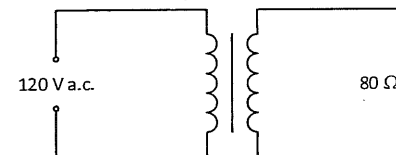


Which of the following correctly describes the magnetic force acting on the rod and the potential of the rod?

	magnetic force on rod	potential across the rod
A	directed to the left	X is higher than Y
B	directed to the right	X is higher than Y
C	directed to the left	Y is higher than X
D	directed to the right	Y is higher than X

22. [H1/H2/Prelims 2018/P1/25]

A non-ideal transformer with an efficiency of 0.75, is connected to a 120 V a.c. supply and a 8.0Ω resistor. The secondary coil of the transformer has twice as many turns as the primary coil.

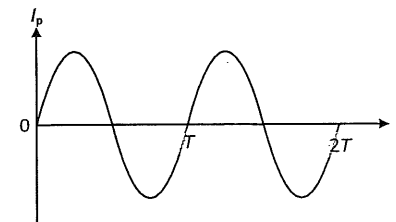


Assuming there is no flux leakage between the primary and secondary coils, what is the current in the primary coil?

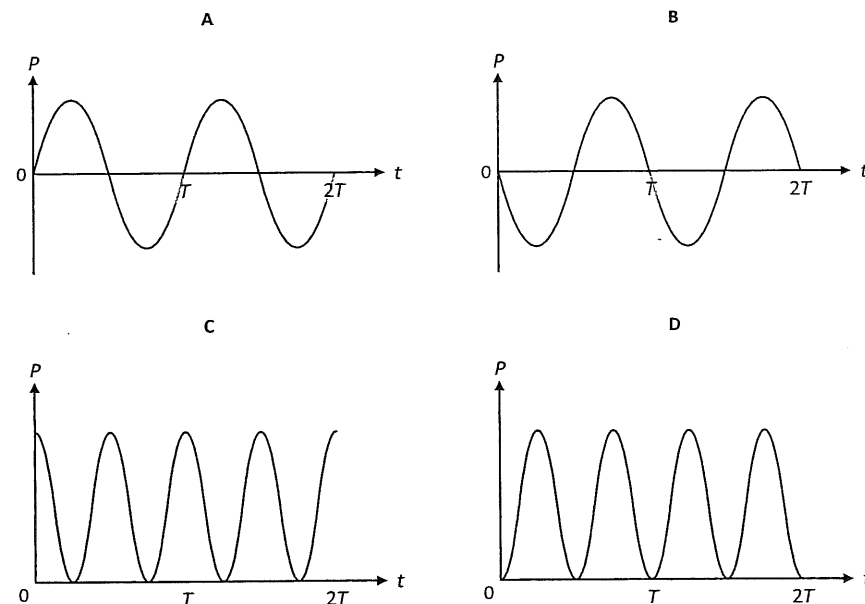
- A 15 A B 30 A C 60 A D 80 A

23. [AJC/H2/Prelims 2018/P1/25 Modified]

The variation with time t of an alternating current I_p connected to the primary coil of a transformer is as shown.

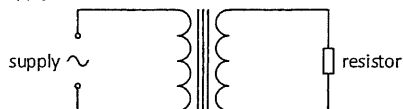


Which graph represents the variation with time t of the power output, P , in a resistor connected across the secondary coil?



24. [AJC/H2/Prelims 2018/P1/26]

A 1:12 step-up ideal transformer is connected to a light bulb. The bulb dissipates heat at a rate of 36 W when the peak alternating voltage of the supply is 2.8 V.



The peak alternating voltage of the supply is then changed to 1.4 V.

What is the root-mean-square current in the primary coil?

- A 6.4 A B 9.1 A C 26 A D 36 A

25. [MJC/H2/Prelims 2018/P1/25]

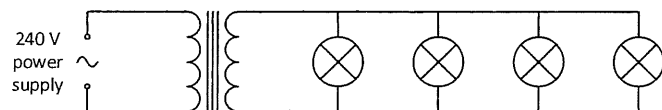
The secondary coil of an ideal transformer delivers an r.m.s. current of 1.5 A to a load resistor of resistance $10\ \Omega$. The r.m.s. current in the primary coil is 5 A.

What is the r.m.s. potential difference across the primary coil?

- A 4.5 V B 6.4 V C 15 V D 50 V

26. [NJC/H2/Prelims 2018/P1/24]

The primary coil of a transformer is connected to a 240 V supply. Four identical lamps are connected to the secondary coil as shown below. The lamps are rated 12 V and 24 W and are operating at normal brightness and the transformer is not 100% efficient.



What could be the possible value of the current that can be drawn from the power supply?

- A 0.10 A B 0.20 A C 0.40 A D 0.45 A

27. [NYJC/H2/Prelims 2018/P1/25 Modified]

An alternating voltage of 230 kV is step down by a transformer, with the secondary output connected to a number of streetlamps with a rating of 230 V 150 W.

If the maximum allowable r.m.s. current through the primary coil is 10 A, what is the maximum number of streetlamps that can be connected in parallel?

- A 1500 B 6500 C 10000 D 15000

II. Structured Questions (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P3/6 Part]

- c. A circular coil is placed with its axis vertical. A bar magnet, aligned with the axis of the coil, is held above the coil as shown in Fig. 1.1. The bar magnet is then dropped from rest. A datalogger connected to the coil records the e.m.f. induced in the coil.

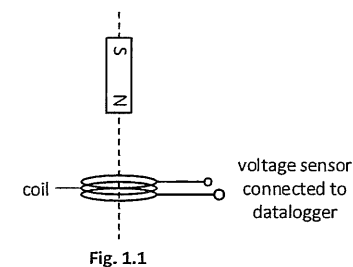


Fig. 1.1

Fig. 1.2 shows the variation with time t of the e.m.f. induced in the coil as the magnet falls through the coil.

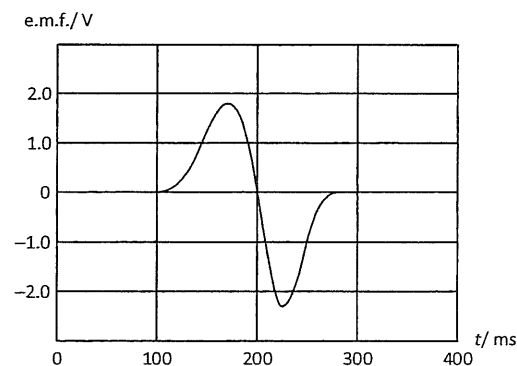


Fig. 1.2

- State what the area under the graph between 200 ms and 275 ms represents. [1]
- Using the laws of electromagnetic induction, explain the shape of the graph obtained in Fig. 1.2. [3]
- The experiment is repeated with the magnet being dropped from a higher height. State and explain how the graph in Fig. 1.2 will change. [2]

[Total: 6]

2. [AJC/H2/Prelims 2018/P3/6 Modified]

A small coil is positioned such that its axis lies along the axis of a large bar magnet, as shown in Fig. 2.1.

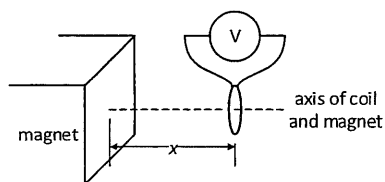


Fig. 2.1

The coil has a diameter of 5.7 mm and contains 180 turns of wire. The ends of the coil are connected to a voltmeter. The variation of the magnetic flux density B through the coil with the distance x between the face of the magnet and the plane of the coil is shown in Fig. 2.2.

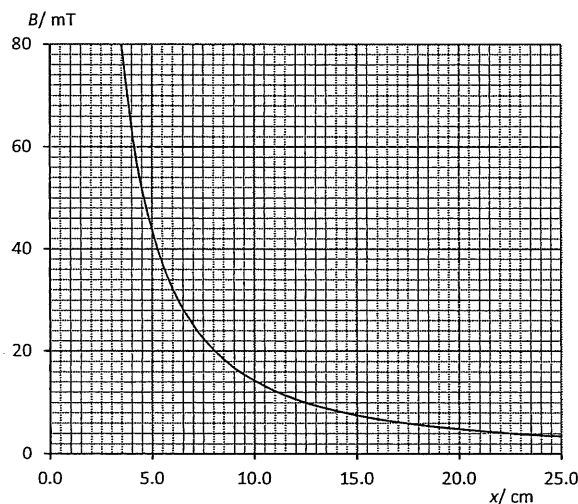


Fig. 2.2

- The coil is initially 5.0 cm from the face of the magnet. Show that the magnetic flux linkage of the coil is 2.0×10^{-4} Wb. [3]
- The coil is then moved along the axis of the magnet so that the distance x increases from 5.0 cm to 20.0 cm in a time of 0.30 s. As the coil is being moved, a deflection is observed in the voltmeter.
 - Determine the average electromotive force (e.m.f.) induced in the coil. [2]
 - State and explain the variation, if any, in the speed of the coil so that the induced e.m.f. remains constant during this movement. [2]
- The voltmeter is now replaced with a resistor and the coil is again moved away from the magnet as in b. As the coil is moved, thermal energy is produced in the resistor. [3]
Use laws of electromagnetic induction to explain the origin of this thermal energy.

[Total: 10]

3. [CJC/H2/Prelims 2018/P2/7]

A spring is attached to the middle of a horizontal wooden rod AB. A U-shaped metal wire ASTB is attached to the rod AB. The U-shaped wire is placed with side ST in a region of uniform magnetic flux that is directed out of the page as shown in Fig. 3.1.

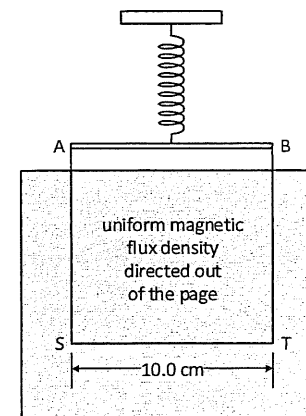


Fig. 3.1

The frame is then pulled down a distance of 1.0 cm and then released. The wire ST undergoes simple harmonic motion in the vertical direction.

- Using Faraday's law of electromagnetic induction, explain why there is an induced e.m.f. in the wire ST while it is in motion. [2]
 - Explain why the induced e.m.f. varies sinusoidally with time. [3]
- The wooden rod AB is replaced with a metal rod. The frame is then set to oscillate.
 - In Fig. 3.2, sketch the variation with time t of the induced e.m.f. across wire ST. Explain your answer. [3]

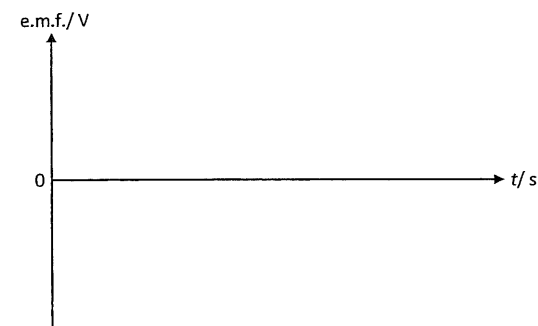


Fig. 3.2

- Explain, how your graph in Fig. 3.2 will change, if any, when the wire frame ASTB is oscillating within the magnetic field. [2]

[Total: 10]

4. [EJC/H2/Prelims 2018/P3/5]

Fig. 4.1 shows the structure of a geophone that is used by geophysicists to determine the speed of seismic waves travelling in the ground.

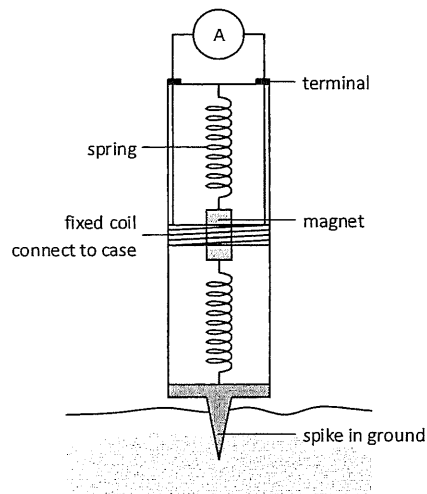


Fig. 4.1

The spike of the geophone is inserted into the ground. When a seismic vibration moves through the case and the coil, the magnet remains stationary due to its inertia.

- State Faraday's law of electromagnetic induction. [1]
- Using Faraday's law, explain how an e.m.f. is generated between the terminals when seismic waves pass through the ground. [2]
- A coil of 50 turns generates a maximum e.m.f. of 85 mV in the geophone. Calculate the rate of change of magnetic flux needed to generate this e.m.f. [2]
- Suggest 2 changes to the geophone which will make it more sensitive to the seismic vibrations. [2]

[Total: 7]

5. [HCI/H2/Prelims 2018/P3/6]

- a. A student writes three incorrect statements as shown in the table below. Each statement has an error either in the unit or number. Circle the error in the statement and write the correct answer.

Incorrect statement	Correct number or unit
The weight of a person is about 700 kg.	
The atmospheric pressure at sea level is about $1.0 \times 10^5 \text{ N m}^2$.	
1 GW is 10 times higher than 1 MW.	

[3]

- b. For over a century, the standard kilogram has been defined by a small platinum-iridium cylinder housed at the International Bureau of Weights and Measures in France. This is the last remaining human-made material object on which a measurement standard is based.

Suggest why the use of a physical sample as the standard became inadequate with the advancement in Science.

[1]

- c. It was proposed at the turn of this century that mass be defined in terms of the Planck constant h rather than in terms of a physical standard mass. This can be done using a watt balance, an electromechanical mass-measuring instrument that measures the mass of a test object very precisely by the strength of an electric current and a voltage. Fig. 5.1 shows a simplified version of the watt balance.

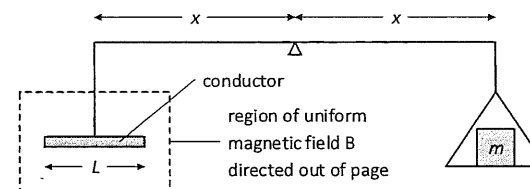


Fig. 5.1

The mass m on the right arm of the balance is the mass to be measured. On the left arm, a conductor of length L is placed in a uniform magnetic field with flux density B pointing out of the page.

The process of finding m involves two stages.

- i. Stage 1: To balance the mass, a constant current I is passed through the conductor such that a magnetic force is exerted on the conductor. Using the principle of moments, the balancing condition is

$$mgx = BILx$$

where g is the gravitational field strength and can be treated as a constant.

Indicate the direction of the conventional current through the conductor by drawing an arrow on it in Fig. 5.1. [1]

- ii. Stage 2: The mass m is removed from the balance and the conductor is made to travel at a constant speed u through the same magnetic field as shown in Fig. 5.2.

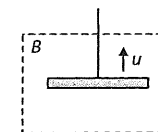


Fig. 5.2

1. Explain how a potential difference is set up across the ends of the conductor. [2]

2. Hence, show that the potential difference V set up is given by BLu . [2]

- *iii. Very precise and accurate values of V and I can be determined using $V = \frac{1}{2} hC/e$ and $I = \frac{1}{2} kne$, where e is the elementary charge, n is an integer and C and k are known constants.

Using c.i. and ii., show how the mass m can be defined in terms of the Planck constant h . [2]

[Total: 11]

6. [H2/Prelims 2018/P3/7 Modified]

- a. State and define the unit for magnetic flux. [2]
- b. Fig. 6.1 shows the top view of a train braking system. A train carriage of mass m is mounted on a rectangular metal frame ABCD of length L and width w , and an effective resistance of R . The train carriage is initially moving at a constant speed along the rails.

A uniform magnetic field B is directed perpendicularly into the ground over a rectangular region of length L . P denotes the start of this region while Q denotes the end of the region. After passing through the magnetic field, the train speed is reduced sufficiently after which brakes can be applied to stop it completely. You may assume that frictional forces are negligible.

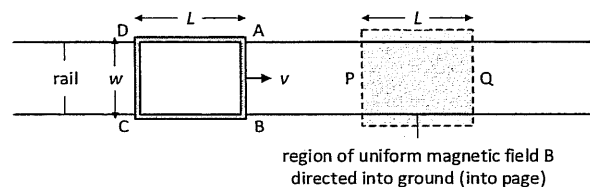


Fig. 6.1 (top view)

- i. Explain how the train is slowed as AB moves through the magnetic field from P to Q. [3]
- ii. Show that the e.m.f. $\mathcal{E}$ induced in the frame is given by Bwv , where v is the speed of the train carriage. Explain your working clearly. [2]
- iii. Hence, deduce an expression for the magnitude of the acceleration of the train carriage as it moves through the magnetic field in terms of B , w , v , m and R . [3]
- c. Fig. 6.2 shows the variation with time of the velocity of the train as it moves through the magnetic field, from the instant AB passes P to the instant CD passes Q.

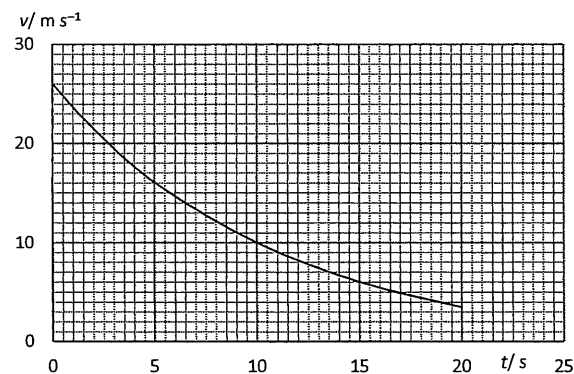


Fig. 6.2

- i. Using Fig. 6.2, estimate the distance PQ. [3]
- ii. Comment on how increasing the region of magnetic field (i.e., distance PQ) would affect the exit speed of the train carriage. [2]
- iii. Sketch on Fig. 6.2 the variation with time of the velocity of the train carriage as it passes through the magnetic field if distance PQ is now reduced. [3]
- iv. Suggest with explanation, one modification such that in the event of a malfunction that results in insufficient deceleration, the train can be slowed down to an acceptable low speed. [2]

[Total: 20]

7. [UC/H2/Prelims 2018/P3/9 Modified]

- a. i. Define the magnetic flux. [1]
- ii. State the laws of electromagnetic induction. [2]
- b. The variation with distance x of the magnetic flux density B beyond one end of a large bar magnet is shown in Fig. 7.1.

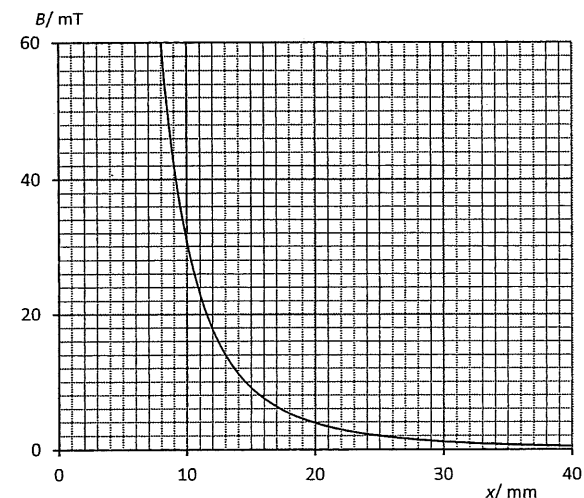


Fig. 7.1

A circular coil of wire with 50 turns and cross-sectional area 16 cm^2 is placed a few centimetres beyond the end of this bar magnet. The axis of the coil is aligned with the axis of the magnet.

- i. Calculate the magnetic flux linkage through the coil when it is 30.0 mm and when it is 10.0 mm from the end of the magnet respectively. [3]
- ii. The coil of wire is moved towards the magnet from the 30.0 mm position to the 10.0 mm position so that a steady e.m.f. of 0.80 mV is induced in the coil. Calculate the average speed of the movement of the coil. [2]
- iii. Describe how the speed of the coil would need to change when moving from the 30.0 mm position to the 10.0 mm position so as to maintain a steady e.m.f. [2]
- c. An a.c. generator consists of a rectangular coil of 800 turns and dimensions $5.0 \text{ cm} \times 8.0 \text{ cm}$ in a uniform magnetic field of magnitude 0.50 T. The coil has a resistance of 0.60Ω and it is connected to an external load of resistance 11.4Ω . The coil is rotating at 240 revolutions per minute.
- i. Calculate the maximum e.m.f. produced by this generator. [2]
- ii. State the relationship between the induced e.m.f. $\mathcal{E}$ and the magnetic flux linkage Φ . [1]

iii. Fig. 7.2 shows the variation with time of the induced e.m.f. E in the coil.

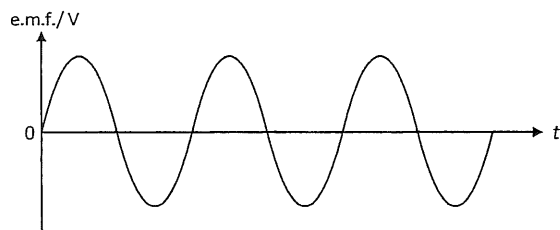


Fig. 7.2

Draw on Fig. 7.2, the corresponding variation of the magnetic flux linkage Φ with time t .

[2]

iv. Calculate the maximum current through the external load.

[1]

v. Explain the terms

1. root-mean-square (r.m.s.) value, and

[1]

2. r.m.s. current.

[1]

vi. Calculate the r.m.s. current through the external load.

[1]

vii. Calculate the average power through the external load.

[1]

[Total: 20]

8. [JJC/H2/Prelims 2018/P3/9 Part Modified]

c. State the laws of electromagnetic induction.

[2]

d. Fig. 8.1 below a square coil GHJK of 500 turns and side 5.0 cm suspended vertically in a uniform horizontal magnetic field of flux density $B = 2.0 \times 10^{-4}$ T. The plane of the coil is at angle $\theta = 35^\circ$ to B .

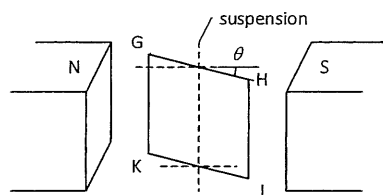


Fig. 8.1

i. Calculate the magnetic flux linkage through the coil.

[2]

ii. The coil is then rotated at 100 Hz about the vertical suspension.

1. Calculate the magnitude of the maximum induced e.m.f. in the coil.

[3]

2. The coil is now acting as an AC generator. It is connected to a heater of resistance 5.00Ω .

Assuming the coil has negligible resistance, calculate the average power generated in the heater.

[3]

3. If the heater in d.ii.2. is also connected in series with an ideal diode, determine the power generated in the heater.

[2]

[Total: 12]

9. [MI/H2/Prelims 2018/P3/8 Modified]

Fig. 9.1 shows a rectangular coil. The coil has 25 turns with dimensions of 15.0 cm by 6.0 cm.

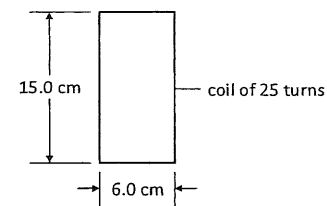


Fig. 9.1

a. i. Define magnetic flux.

[1]

ii. A uniform magnetic field of flux density 25 mT is at right angles to the plane of the coil initially. Calculate the magnetic flux through the coil at this instance.

[1]

b. i. Fig. 9.2 shows the sinusoidal variation with time t of the magnetic flux density B through the rectangular coil in Fig. 9.1.

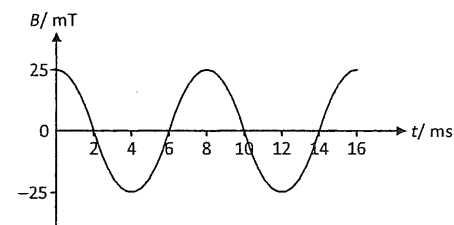


Fig. 9.2

Using Faraday's Law, explain why the variation in magnetic flux density through the coil as shown in Fig. 9.2 leads to a generation of sinusoidal alternating e.m.f.

[3]

ii. Show that the equation for the induced e.m.f. of the coil is

$$E = NB_0 A \cdot \frac{2\pi}{T} \cdot \sin\left(\frac{2\pi}{T} \cdot t\right)$$

where N is the number of turns in the coil, A is the area of coil, B_0 is the maximum magnetic flux density and T is the period of the change in magnetic flux density.

[2]

iii. Hence, determine the maximum magnitude of the induced e.m.f. of the coil.

[2]

iv. On Fig. 9.3, sketch a graph to show the variation with time of the e.m.f. induced in the coil from 0 ms to 16 ms.

[2]

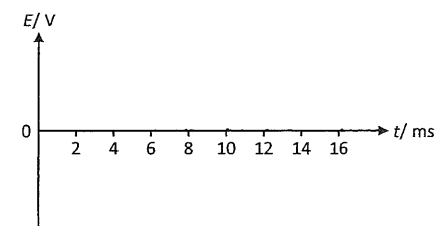


Fig. 9.3

- c. i. The AC generated by the rectangular coil passes through a transformer. The primary coil of the ideal transformer has 15 turns.
Assuming that the transformer is ideal, calculate the number of turns in the secondary coil if the value of root-mean-square potential difference of the output at the secondary coil is 240 V. [3]
- ii. A non-ideal transformer has an input e.m.f. of 12 V while the secondary coil draws a current of 0.25 A with a potential difference of 240 V.
If the efficiency of the transformer is 81 %, determine the current in the primary coil. [2]
- d. In many distribution systems for electrical energy, the power is transmitted using alternating current at high voltages. Suggest and explain an advantage, one in each case, for the use of
- alternating voltages,
 - high voltages.

[Total: 20]

10. [MJC/H2/Prelims 2018/P2/5]

- a. Define magnetic flux. [1]
- b. Fig. 10.1 shows a 1.6 m long solenoid with 400 turns and a cross-sectional diameter of 4.0 cm. A coil Y, with 80 turns, is wound tightly around the centre region of the solenoid.

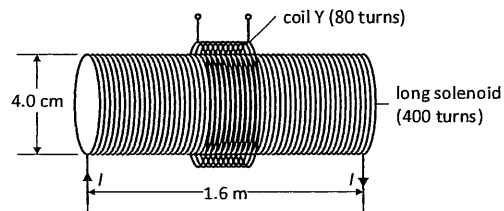


Fig. 10.1

- i. Show that, for a current $I = 3.8$ A in the solenoid, the magnetic flux linkage of coil Y is 1.2×10^{-4} Wb. [2]
- ii. The current I in the solenoid in b.i. is reversed in 0.30 s. Calculate the mean e.m.f. induced in coil Y. [2]
- iii. The variation with time of the current I in the solenoid in b.ii. is shown in Fig. 10.2.

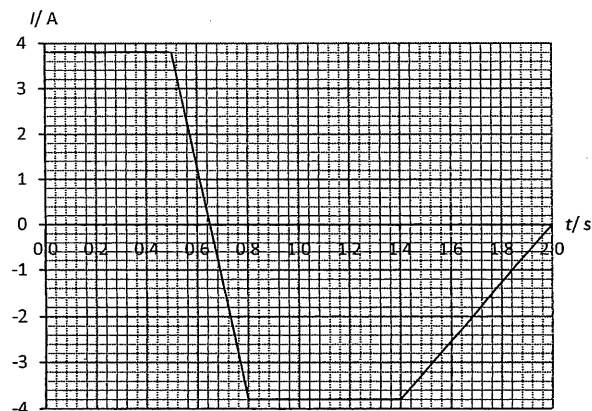


Fig. 10.2

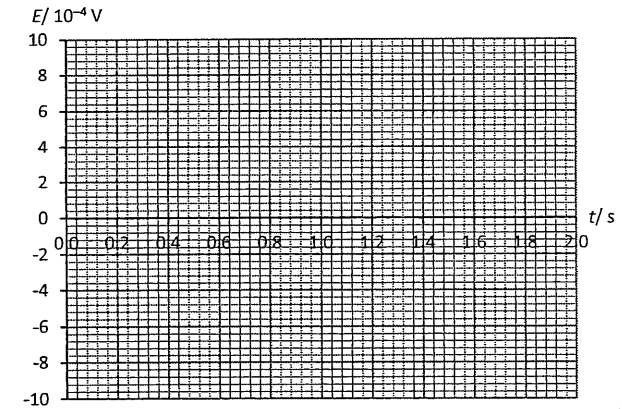


Fig. 10.3

Use your answer to b.ii. to sketch, on Fig. 10.3, the variation with time t of the e.m.f. E induced in coil Y. [3]

- iv. An iron core is inserted into the solenoid and then held stationary within the solenoid. Explain the effect on the e.m.f. induced in coil Y. [2]

[Total: 10]

11. [NYJC/H2/Prelims 2018/P3/6 Modified]

- a. State Lenz's Law. [1]
- b. A strong magnet falls through a solenoid which is wound around a cardboard tube. The solenoid is connected to a switch and an ammeter as shown in Fig. 11.1.

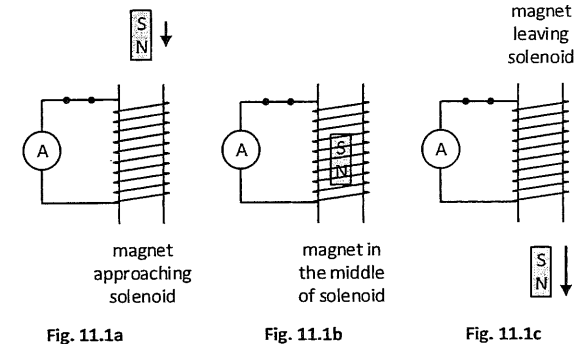


Fig. 11.1a

Fig. 11.1b

Fig. 11.1c

- i. On Fig. 11.1a, Fig. 11.1b and Fig. 11.1c, indicate with arrows on the windings of the solenoid the direction of any induced current. [2]
- ii. Hence, on Fig. 11.2, sketch a labelled graph to show the variation with time t of the current I induced in the solenoid as the magnet falls through it. [2]

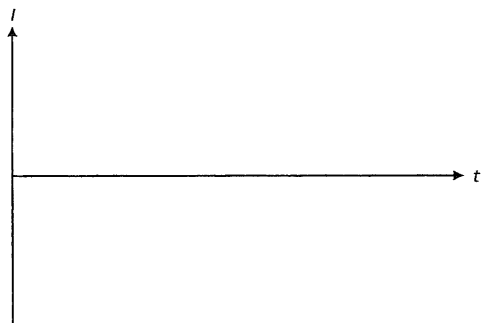


Fig. 11.2

- iii. The switch connected to the solenoid is now opened, and the same magnet is allowed to drop through the solenoid again. State and explain any change in the speed of the magnet as it leaves the solenoid. [2]

[Total: 7]

12. [PIC/H2/Prelims 2018/P3/7]

- a. Define *magnetic flux*. [1]
- b. Fig. 12.1 shows two short coils connected in series by a switch S.

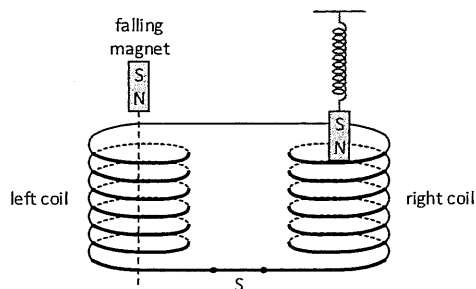


Fig. 12.1

- i. Describe and explain the behaviour of the suspended magnet above the right coil after a bar magnet is dropped through the left coil. [4]
- ii. Switch S is now opened. State and explain any changes to your answer in b.i. [2]

[Total: 7]

13. [RVHS/H2/Prelims 2018/P3/8]

A typical three-phase generator in turbines use for power generation consists of a rotating electromagnet surrounded by three sets of stationary wire coils AA', BB' and CC', with angles of 120° between them, wrapped around metal cores to generate electricity, as shown in Fig. 13.1.

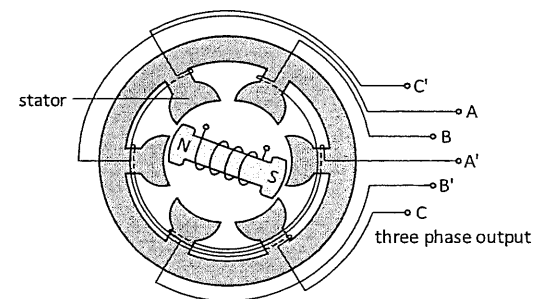


Fig. 13.1

Each set of coil (AA', BB' and CC') is a solenoid that consists of wires tightly wound around a rigid metal core as shown in Fig. 13.2.

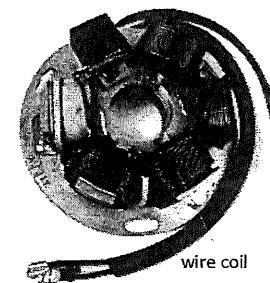


Fig. 13.2

The electromagnet shown in Fig. 13.1 can rotate about the axis perpendicular to the plane of the page, and is connected to turbine blades that rotate the electromagnet at a constant rate such that electricity can be generated in the coils.

- a. State Faraday's law of electromagnetic induction. [2]
- b. i. Ignoring coils BB' and CC', explain why the rotation of the electromagnet causes a sinusoidal e.m.f. to be induced in coil AA'. [3]

ii.

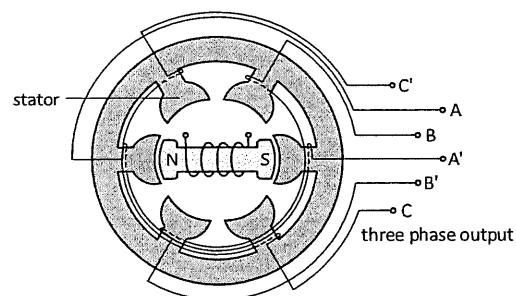


Fig. 13.3

At a particular instant of time $t = 0$ s, the magnet is aligned with the coil labelled AA' as shown in Fig. 13.3. Coil AA' has a total of 500 turns with a diameter of 0.10 m. The electromagnet generates an effective uniform magnetic flux density of 0.80 T at the position of the coils when its axis aligns with that of the coils.

Calculate the magnetic flux linkage of coil AA' at this instant. Give an appropriate unit. [3]

- iii. The variation with time of magnetic flux linkage of coil AA' is as shown in Fig. 13.4. The flux linkage varies sinusoidally with time.

When operating optimally, the magnet makes 1800 revolutions per minute.

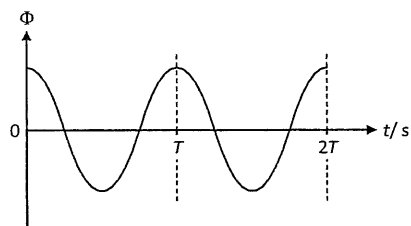


Fig. 13.4

Write an expression for the magnetic flux linkage Φ in the coil as a function of time t as shown in Fig. 13.4. [1]

- iv. Hence, show that the maximum induced e.m.f. in the coil is 590 V. [2]
- v. On Fig. 13.5, sketch the corresponding graph of the variation with time of induced e.m.f. in the coil AA'. Explain, with reference to Fig. 13.4, the position of maximum induced e.m.f. in your graph. [3]

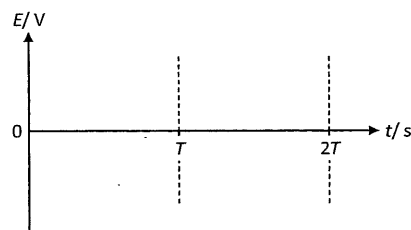


Fig. 13.5

- c. Two additional sets of coils BB' and CC' complete the three-phase generator as shown in Fig. 13.3. As the electromagnet rotates anti-clockwise, the induced e.m.f. in coils AA', BB' and CC' are $\frac{2}{3}\pi$ rad out of phase with one another.

On Fig. 13.5, sketch additional graphs to show how the induced e.m.f. in coils BB' and CC' vary with time. Label your sketches clearly. [2]

- d. The output of the generator is connected to the input of an ideal transformer as shown in Fig. 13.6.

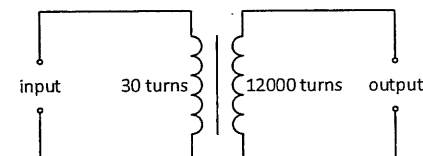


Fig. 13.6

The peak input current is measured to be 20 A. The transformer has 30 turns on its primary coil and 12000 turns on its secondary coil. The output in the secondary coil circuit is connected to transmission cables used for power transmission to homes more than 20 km away.

- i. Explain what is meant by an ideal transformer. [1]
- ii. Determine the r.m.s. current in the secondary coil. [2]
- iii. Hence, or otherwise, suggest why the e.m.f. generated is stepped up to high voltages for long distance transmission. [1]

[Total: 20]

14. [VJC/H2/Prelims 2018/P2/5 Modified]

- a. State Lenz's law, and explain how you would use a coil and a magnet to demonstrate the law. Make clear the use of any other apparatus in your explanation. [4]
- b. An instrument based on induced electromotive force (e.m.f.) is used to measure the speed of a projectile. A small, strong but light magnet is imbedded in the projectile as shown in Fig. 14.1. The projectile passes at a constant speed through the two identical coils with N turns each, separated by a distance d between them. As the projectile passes through each coil, a pulse of e.m.f. is induced in the coil. The time interval between the pulses can be measured accurately with an oscilloscope.

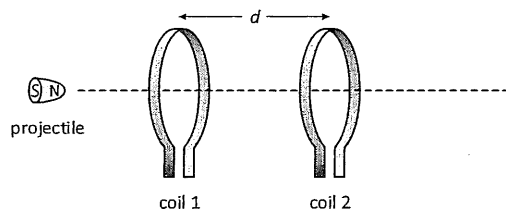


Fig. 14.1

- i. Use the laws of electromagnetic induction to explain why an e.m.f. is induced in the coils. [2]
- ii. In Fig. 14.2, draw clearly a graph of the variation with time of induced e.m.f. E for the arrangement shown in Fig. 14.1.

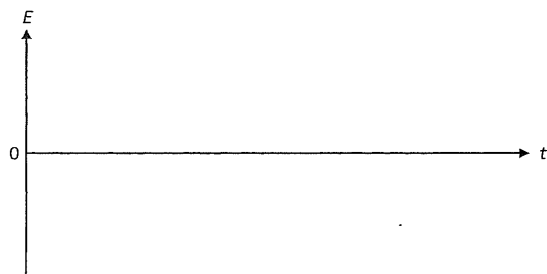


Fig. 14.2

- On your graph, label clearly which pulse is from coil 1 and which is from coil 2. [2]
- iii. If the pulse separation is 2.40 ms and d is 1.50 m, calculate the speed of the projectile. [1]
- iv. State and explain how the graph in Fig. 14.2 will change if the speed of the projectile is increased. [2]

[Total: 11]

15. [YJC/H2/Prelims 2018/P3/6 Part Modified]

- d. A student sets up the apparatus shown in Fig. 15.2a and Fig. 15.2b to determine the acceleration of a cart down a ramp. Fig. 15.2a shows the setup from the front view and Fig. 15.2b shows the side view of the set-up. As the cart passes the coils which are suspended, an e.m.f. is induced in each coil. The outputs of the coils are monitored using the voltage sensors connected to a datalogger and computer. The voltage sensors, datalogger and computer are not shown in the figures. Fig. 15.3 shows the computer printouts after one test.

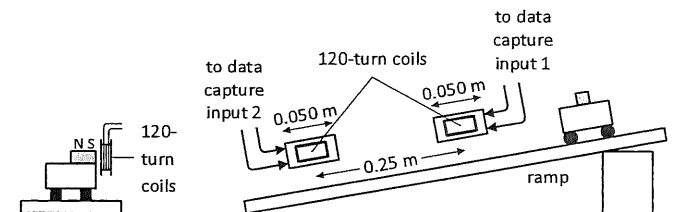


Fig. 15.2a (Front View)

Fig. 15.2b (Side View)

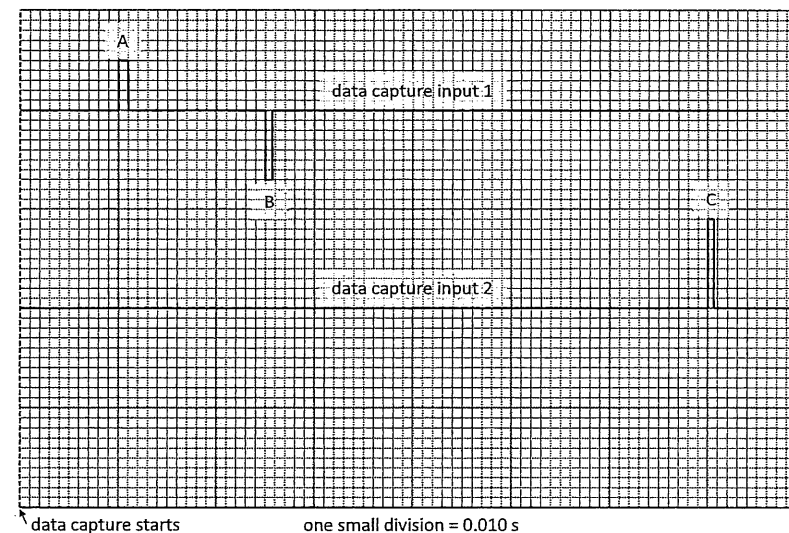


Fig. 15.3

Explain why

- i. e.m.f. is induced in both coils, [1]
- ii. pulse B is in the opposite direction to pulse A, [2]
- iii. the amplitude of pulse C is larger than pulse A. [2]

[Total: 5]

16. [DHS/H2/Prelims 2018/P3/6]

- a. An alternating voltage of period 10 ms is being applied directly across a resistor of $5.0\ \Omega$ in a circuit. The variation with time t of voltage V is shown in Fig. 16.1.

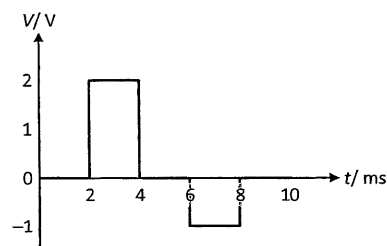


Fig. 16.1

Calculate the steady voltage passing through the same resistor that would produce an identical heating effect. [2]

- b. Explain why it is necessary to use high voltages for the efficient transmission of electrical energy [2]
- c. Another sinusoidal voltage input of 6.5 mV r.m.s. and 50 Hz is now connected to the primary coil of a transformer as shown in Fig. 16.2. The transformer is assumed to be ideal and its turns ratio, $\frac{N_s}{N_p} = 71$.

The secondary coil is connected to a resistor R . An average power of 0.040 W is produced in resistor R .

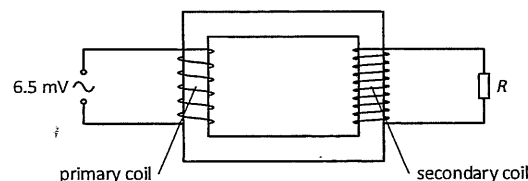


Fig. 16.2

- i. Calculate the r.m.s output voltage supplied to resistor R . [1]
- ii. In Fig. 16.3, sketch the variation with time t of the power P dissipated in the resistor R . Label all values on the axes. [2]

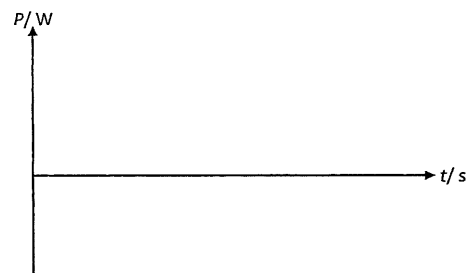


Fig. 16.3

- iii. An ideal diode is now connected to the secondary coil with resistor R as shown in Fig. 16.4.

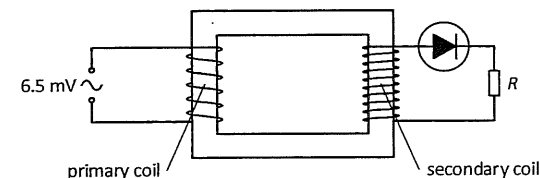


Fig. 16.4

Describe the variation with time of the

- current flow through resistor R , and [2]
- voltage across resistor R . [1]

[Total: 10]

17. [NYJC/H2/Prelims 2018/P2/8]

In Singapore, the power stations are situated near the coast because sea water is used to condense steam into water. The transmission of electric power over long distances from the power stations to homes would not be feasible without transformers.

A step-up transformer near the power station boosts the station's output root-mean-square (r.m.s.) voltage from 12.0 kV to 240 kV and a series of step-down transformers near the homes reduces the r.m.s. voltage to a final value of 240 V at the homes. The transmission voltage has a frequency of 50 Hz.

- a. For the transformer located near the power station, determine the ratio

$$\frac{N_p}{N_s} = \frac{\text{number of turns of primary coil}}{\text{number of turns of secondary coil}}$$

[1]

- b. Explain why

- the voltage is stepped up near the power plant, [2]
 - an alternating voltage is used instead of direct voltage to transmit electrical energy [1]
- c. i. Determine the r.m.s. current delivered to an electric kettle rated at 700 W when it is connected to a power socket in the house. [1]
- ii. Calculate the peak output voltage to the electric kettle. [1]
- d. On Fig. 17.1 below, sketch a labelled graph for the variation with time t of the thermal power P dissipated by the electric kettle. [2]

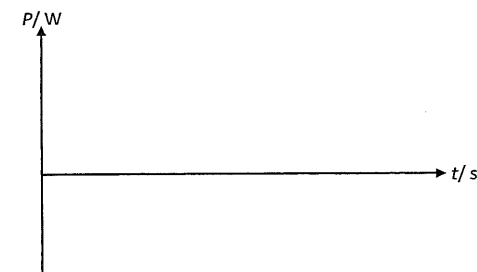


Fig. 17.1

[Total: 8]

18. [SRJC/H2/Prelims 2018/P3/5]

An ideal transformer is connected to a sinusoidal a.c. supply, as shown in Fig. 18.1. The primary coil has a r.m.s. current of 0.85 A.

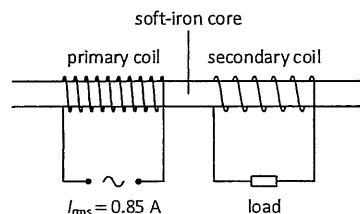


Fig. 18.1

- a. Use the laws of electromagnetic induction to explain how a potential difference can be developed across the secondary coil. [3]
- b. i. The primary coil contains 9 turns per cm, calculate the maximum magnetic flux density at its centre. [2]
 ii. The ratio of the number of turns in the primary to secondary coil is 16:1. Calculate the peak current in the load. [2]
 iii. The variation with time t of current I_p in the primary coil is shown in Fig. 18.2. On Fig. 18.3, sketch the variation with time t of current I_s in the secondary coil.

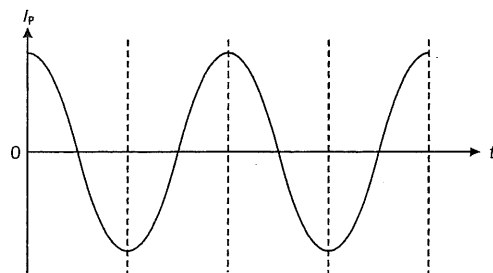


Fig. 18.2

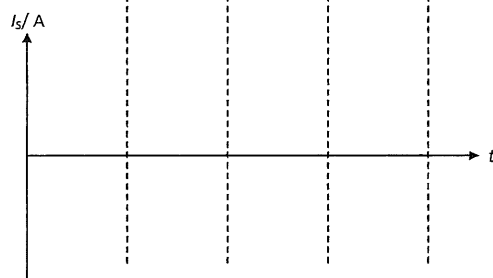


Fig. 18.3

[1]

- iv. State and explain how the answer to b.iii. will change if the iron core is removed from the transformer. [2]

[Total: 9]

19. [DHS/H2/Prelims 2018/P3/6]

- a. An alternating voltage of period 10 ms is being applied directly across a resistor of 5.0Ω in a circuit. The variation with time t of voltage V is shown in Fig. 2.1.

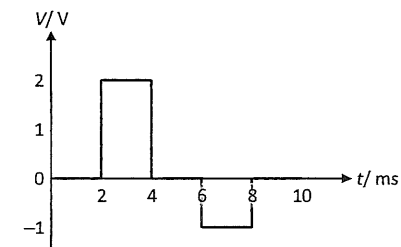


Fig. 2.1

Calculate the steady voltage passing through the same resistor that would produce an identical heating effect. [2]

- b. Explain why it is necessary to use high voltages for the efficient transmission of electrical energy [2]
- c. Another sinusoidal voltage input of 6.5 mV r.m.s. and 50 Hz is now connected to the primary coil of a transformer as shown in Fig. 2.2. The transformer is assumed to be ideal and its turns ratio, $\frac{N_s}{N_p} = 71$.

The secondary coil is connected to a resistor R . An average power of 0.040 W is produced in resistor R .

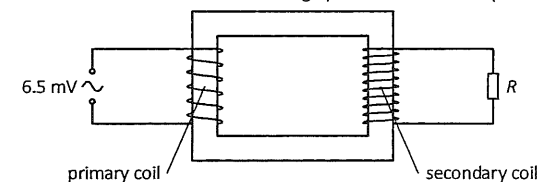


Fig. 2.2

- i. Calculate the r.m.s output voltage supplied to resistor R . [1]
- ii. In Fig. 2.3, sketch the variation with time t of the power P dissipated in the resistor R . Label all values on the axes. [2]



Fig. 2.3

- iii. An ideal diode is now connected to the secondary coil with resistor R as shown in Fig. 2.4.

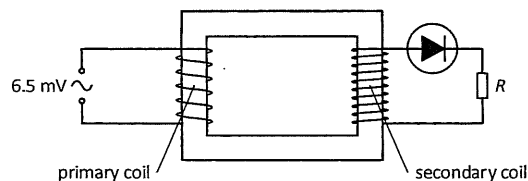


Fig. 2.4

Describe the variation with time of the

1. current flow through resistor R , and
2. voltage across resistor R .

[2]

[1]

[Total: 10]

20. [EJC/H2/Prelims 2018/P3/6]

In the hairdryer shown in Fig. 3.1, an alternating current passes through the heating coil when the switch is closed.

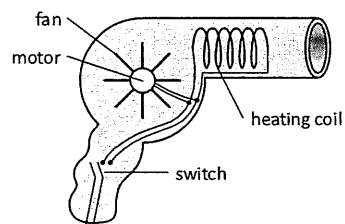


Fig. 3.1

- a. By reference to heating effect, explain what is meant by the root-mean-square (r.m.s) value of an alternating current. [1]
- b. The hair dryer is connected to the mains supply of 120 V r.m.s. and frequency 50 Hz. The heating coil delivers a power of 1200 W.
 - i. Calculate the peak current through the heating coil. [2]
 - ii. Write an equation, in terms of the elapsed time t , for the current that passes through the heating coil, given that the instantaneous power output is zero at $t=0$ s. [1]
- c. The primary coil of a transformer is connected to a 2.4 kV r.m.s. supply. The secondary coil is connected to the hair dryer and the current flowing through the heating coil has the same value as that calculated in b.i.. The transformer is non-ideal and electrical energy is converted to thermal energy in the transformer at a rate of 600 W. Determine the r.m.s. current in the primary. [2]

[Total: 6]

21. [NYJC/H2/Prelims 2018/P2/8]

In Singapore, the power stations are situated near the coast because sea water is used to condense steam into water. The transmission of electric power over long distances from the power stations to homes would not be feasible without transformers.

A step-up transformer near the power station boosts the station's output root-mean-square (r.m.s.) voltage from 12.0 kV to 240 kV and a series of step-down transformers near the homes reduces the r.m.s. voltage to a final value of 240 V at the homes. The transmission voltage has a frequency of 50 Hz.

- a. For the transformer located near the power station, determine the ratio

$$\frac{N_p}{N_s} = \frac{\text{number of turns of primary coil}}{\text{number of turns of secondary coil}}$$
 [1]
- b. Explain why
 - i. the voltage is stepped up near the power plant, [2]
 - ii. an alternating voltage is used instead of direct voltage to transmit electrical energy [1]
- c. i. Determine the r.m.s. current delivered to an electric kettle rated at 700 W when it is connected to a power socket in the house. [1]
 - ii. Calculate the peak output voltage to the electric kettle. [1]
- d. On Fig. 4.1 below, sketch a labelled graph for the variation with time t of the thermal power P dissipated by the electric kettle. [2]

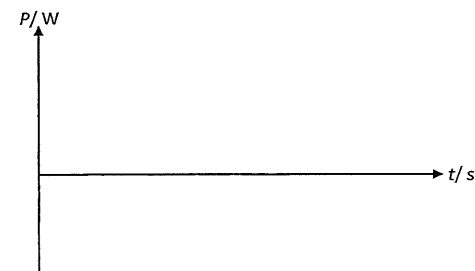


Fig. 4.1

[Total: 8]

22. [SRJC/H2/Prelims 2018/P3/5]

An ideal transformer is connected to a sinusoidal a.c. supply, as shown in Fig. 6.1. The primary coil has a r.m.s. current of 0.85 A.

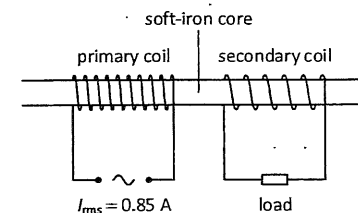


Fig. 6.1

- a. Use the laws of electromagnetic induction to explain how a potential difference can be developed across the secondary coil. [3]
- b. i. The primary coil contains 9 turns per cm, calculate the maximum magnetic flux density at its centre. [2]
 - ii. The ratio of the number of turns in the primary to secondary coil is 16:1. Calculate the peak current in the load. [2]

- iii. The variation with time t of current I_p in the primary coil is shown in Fig. 6.2.

On Fig. 6.3, sketch the variation with time t of current I_s in the secondary coil. [1]

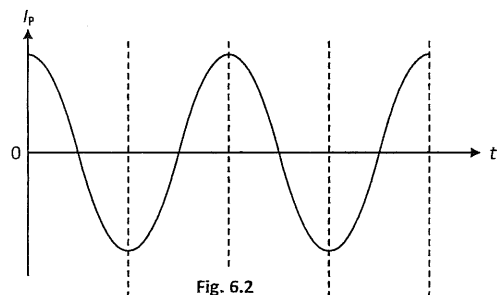


Fig. 6.2

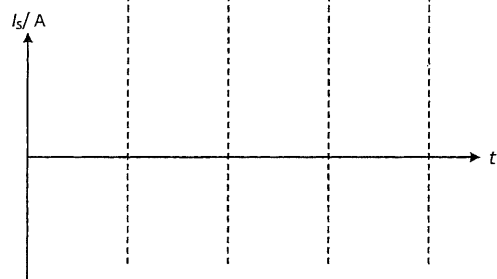


Fig. 6.3

- iv. State and explain how the answer to b.iii. will change if the iron core is removed from the transformer. [2]